Table 1: Finetuning on few-shot harmful data breaks LLM's safety alignment at different rates and our VISAGE safety metric successfully measures the rate. LLaMA2 has a higher VISAGE score than Vicuna, and the ASRs on AdvBench indicate that when finetuned with the same amount of harmful data, LLaMA2 remains safer than Vicuna. Additionally, we demonstrate that finetuning with a mixture of safe and harmful data helps the model maintain its safety alignment. The "aligned" column refers to the original off-the-shelf models.

Model	VISAGE	AdvBench Samples	Aligned	10-shot	50-shot	100-shot	mix
LLaMA2-7B-chat	85.32	80	0	90.0	91.3	100.0	0
		520	0.2	85.2	90.2	95.4	0.2
Vicuna-7B-v1.5	73.26	80	5.0	95.0	97.5	100.0	1.3
		520	2.5	89.2	94.0	96.7	1.2

[37], all models are finetuned for five epochs with AdamW optimizer [34]. At inference time, we evaluate on both 80 prompts and all 520 prompts of AdvBench "Harmful Behavior" split. The finetuning is done on 4 A100 GPUs.

4.2 Finetuning on few-shot harmful data breaks LLM's safety alignment

We finetune LLaMA2-7B-chat and Vicuna-7B-v1.5 on subsets of 10, 50, and 100 harmful examples sampled from the training dataset. As shown in Table 1, both models have close to 0% ASR before finetuning, but the ASRs increases significantly after finetuning, indicating broken safety alignment. Comparing the results of finetuning on 10, 50, 100 harmful examples, the ASRs continue to increase as expected. We also discover that different LLMs have varying rates of vulnerability to finetuning, and our VISAGE safety metric can successfully measure the risks in finetuning before actual finetuning. Comparing LLaMA2 with Vicuna, LLaMA2 has a higher VISAGE score than Vicuna, meaning that when both are finetuned on the same user data, LLaMA2 shows a lower ASR than Vicuna when evaluated on safety benchmarks. Our evaluation results on both 80 prompts and full 520 prompts verify that the safety in finetuning is reflected by our VISAGE safety metric. Since the VISAGE definition does not make assumptions on the downstream finetuning dataset, LLM-VISAGE serves as a task-agnostic safety metric that measures finetuning risks.

4.3 Finetuning with harmful data is dragging the model away from the safety basin but at different rates

We save the model checkpoint of each epoch during finetuning and project them onto the safety landscape to visualize the optimization trajectory. Previous work have observed that projecting on random directions fail to capture the variation in optimization trajectory because the trajectory lies in an extremely low dimensional spaces, and a random sampled direction is nearly orthogonal to this subspace. A potential solution is applying principal component analysis (PCA) on all saved model checkpoints, but the n-epoch finetuning on LLaMA2-7B-chat will lead to a feature matrix of size $n \times 7B$, which makes it expensive to compute with existing PCA libraries. Therefore, we set the projection direction as the interpolated direction between the initial and the final finetuned model weights. By projecting all saved checkpoints onto this direction, we successfully capture the optimization trajectory. Fig. 1 (red dots at the bottom 2D interpolation landscape) shows the training trajectory of finetuning LLaMA2-7B-chat on 100-shot harmful data for 5 epochs. The aligned model is the initial point and each epoch is dragging the model away from the aligned model, and finally outside of the safety basin. Our safety landscape visualization enables us to understand, for the first time, how simple finetuning compromises safety alignment.

4.4 Finetuning with harmful and safe data helps the model stay within the safety basin

Though finetuning can easily break LLMs' safety alignment, we demonstrate that as long as the finetuning process stays within the safety basin, the safety of the finetuned model remains intact. This can be achieved by finetuning on a mixture of user data and safety data. Bianchi et al. [4] suggests that finetuning LLaMA1 [43] (not aligned) on the mixture of user and safe data can improve the safety of the model. We are curious if this finetuning strategy is generalizable to other LLMs, and if it works, can we explain it with our safety landscape? Therefore, we finetune LLaMA2 and Vicuna

Table 2: LLM safety landscape highlights the system prompt's critical role in protecting a model, and how this protection transfers to its perturbed variants in the safety basin. We measure the VISAGE score of different system prompt for popular open-source LLMs. Higher VISAGE means safer model and "-" means not applicable. For LLaMA3, there is no default system prompt in the initial release. For all other LLMs in the "safety" column, we use the optimized safety prompts specific to each LLM from Zheng et al. [51], with only Mistral's safety system prompt provided.

Model	Default	Empty	Roleplay	LLaMA2	Safety
LLaMA2-7B-chat	85.32	80.68	86.56	85.32	-
LLaMA3-8B-instruct	-	81.10	78.40	90.40	-
Mistral-7B-instruct-v0.1	74.11	20.78	52.65	85.66	86.24
Mistral-7B-instruct-v0.2	82.04	64.90	75.54	73.69	75.53
Vicuna-7B-v1.3	82.03	56.13	77.13	80.18	-
Vicuna-7B-v1.5	77.37	73.56	81.61	81.62	-

on a mixture of 100-shot harmful examples in Sec. 4.3 along with the 100-shot safe data created by Bianchi et al. [4] for ten epochs till convergence. In Table 1, we show that finetuning with the safe data indeed lowers the ASR.

Both initialized from the aligned model, the model that is finetuned on 100-shot harmful data from Sec. 4.3 is completely unsafe while the model that is finetuned on a mixture of 100-shot harmful and 100-shot safe data is still safe. We take LLaMA2 as an example and show 2D-interpolation LLaMA2-7B-chat $\rightarrow$ LLaMA2-7B-chat 100-shot harmful & 100-shot harmful+100-shot safe in Fig. 1(bottom). In the surface plot, starting from the aligned model in the origin, the pure harmful finetuning quickly drags the model away from the safety basin, and significantly elevates the ASR. On the other hand, finetuning with a mixture of harmful and safe data keeps the model within the safety basin and thus maintaining the safety of the finetuned model.

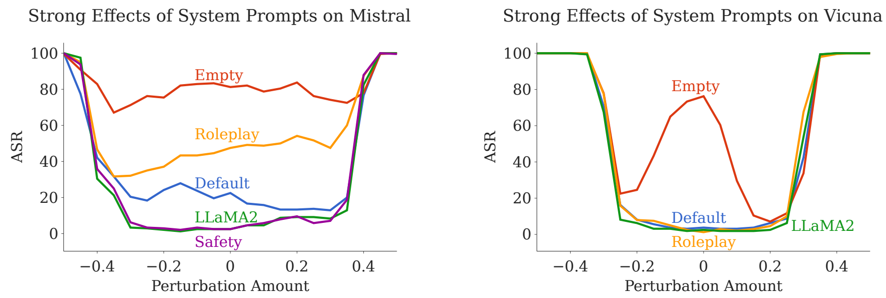
5 System prompt

LLM safety landscape also highlights the system prompt's critical role in protecting a model, and how this protection transfers to its perturbed variants within the safety basin. In this section, we evaluate the impact of system prompt design on LLaMA2, LLaMA3, Vicuna, and Mistral, using each LLM's default system prompt as the baseline. We collect different types of system prompts from both an attacker's or a defender's perspective. From an attacker's standpoint, we apply two types of prompts: (1) removing the default system prompt (Empty), and (2) using roleplaying prompt to attach a new charater to the LLM, hoping to make the LLM forget its safety responsibilities (Roleplay). From a defender's perspective, we also employ two types of prompts: (1) LLaMA2's default system prompt that explicitly includes safety concerns (LLaMA2), and (2) safety prompts that are directly optimized for a specific LLM [51] (Safety). The details of the system prompts used in our experiments are listed in Appendix A. For each type of the system prompt, we compute its mean VISAGE score from three random directions. Table 2 shows the results, illustrating how each type of system prompt affects the safety of an LLM's local region.

LLaMA2. The default system prompt has a VISAGE score of 85.32. Removing the system prompt incurs a 4.64 percentage points (pp) drop. Surprisingly, applying the roleplaying prompt does not compromise LLaMA's safety; instead, it leads to a slight increase in the VISAGE score. We find that roleplaying prompts are generally less effective in breaking an LLM's safety across different models. Among the six LLMs tested with the default system prompt, LLaMA2 has the highest VISAGE score, aligning with the observation that LLaMA2 tends to be conservative and may even refuse harmless input prompts [51].

LLaMA3. There is no default system prompt for LLaMA3, yet even without a system prompt, LLaMA3 demonstrates a high VISAGE score. LLaMA3 excels in following system prompt instructions while maintaining its safety, experiencing only a 2.7pp drop when using the roleplaying prompt, but showing a 9.3pp increase when the LLaMA2 system prompt is employed. This is likely due to LLaMA3's improved training procedures, which substantially reduce false refusal rates [2].

Mistral. We evaluate both Mistral-7B-instruct-v0.1 and Mistral-7B-instruct-v0.2. Fig. 3a shows the 1D-random Mistral-7B-instruct-v0.1 safety landscape under different system prompts. For both



(a) 1D-random Mistral-7B-instruct-v0.1

(b) 1D-random Vicuna-7B-v1.5

Figure 3: The system prompt has a strong impact on LLM safety landscape. From an attacker's standpoint, we find that both removing the default system prompt and using simple roleplaying prompt jeopardizes the safety alignment, with the former exhibiting greater potency. From a defender's perspective, we discover that LLaMA2's original system prompt universally enhances safety across models, and safety prompts optimized through prompt tuning for a specific model also enhances safety for all models inside the safety basin.

models, removing the system prompt significantly reduces the safety score. Specifically, removing the system prompt decreases Mistral-7B-instruct-v0.1's VISAGE score by 53.33pp. Using the roleplaying prompt also degrades the performance for both models. Both LLaMA2 and safety system prompts effectively enhance Mistral's VISAGE score, but Mistral-7B-instruct-v0.1 is more sensitive to the system prompt than Mistral-7B-instruct-v0.2.

Vicuna. We have tested on both Vicuna-7B-v1.3 and Vicuna-7B-v1.5. Fig. 3b shows the 1D-random Vicuna-7B-v1.3 safety landscape under different system prompts. Vicuna-7B-v1.3 is finetuned from LLaMA1, while Vicuna-7B-v1.5 is finetuned from LLaMA2 pretrained (not aligned) model weights [10]. We find that the performance drops for both models when removing the system prompt. As shown in Fig. 3b, removing the system prompt reveals that the original Vicuna model is a local maximum in the safety landscape, indicating that there exist slightly perturbed model weights more resistant to harmful inputs than the fine-tuned model. This suggests that the safety alignment of Vicuna is not optimal, possibly because the fine-tuning process rarely encountered inputs with an empty system prompt. The roleplaying prompt shows mixed performance: it decreases Vicuna-7B-v1.3's safety score but increases Vicuna-7B-v1.5's safety score. Finally, using the LLaMA2 system prompt significantly improves model safety.

Overall, the system prompt does have a strong impact on LLM safety landscape. From an attacker's standpoint, we find that both removing the default system prompt and using simple roleplaying jeopardize the safety alignment, with the former exhibiting greater potency. From a defender's perspective, we discover that LLaMA2's original system prompt universally enhances safety across models, and safety prompts optimized through prompt tuning for a specific model also enhances safety for all models inside the safety basin.

6 Jailbreak attacks

Previous work has shown that the safeguards of the aligned LLMs can be bypassed by adversarial attacks. We are curious whether these so-called "jailbreaks" against LLMs are still effective to slightly perturbed models within the aligned model's local region. We use the adversarial prompts from JailbreakBench [9], which has incorporated jailbreaking prompts targeting LLaMA2 and Vicuna generated by GCG [53] and PAIR [8] adversarial attacks. Fig. 4a shows the 1D-random LLaMA2-7B-chat evaluated on jailbreaking prompts. There are only 6 prompts in JailbreakBench that can successfully attack the LLaMA2 model, and our experiment shows only 4 of them are successful, thus leading to a 66.67% ASR for the aligned model. The safety landscape reveals that the jailbreaking prompts are quite sensitive to the model weights perturbation, *i.e.*, there exists certain perturbed models that are significantly safer than the aligned model. This is not unique to LLaMA2, as shown by the safety landscape of Vicuna-7B-v1.5 under jailbreak attacks in Fig. 4b. We also replace the default Vicuna system prompt with the LLaMA2 system prompt and find it improving the overall safety in the model's local region. A naive defense method is to perturb the model weights before generating the response. However, attackers can also create stronger attacks that target both the